Critical Infrastructure Security

IY5612 (IY3612/IY4612) — Academic Year 2025/2026 (Unit 4)

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Unit 4— Critical Infrastructures and Interdependency

Contents

Internet Robustness

Internet Robustness
- Global Internet Structure
- Network Robustness

Robustness of the Internet Infrastructure

Economic Models of Interdependency

Part I

Internet Robustness

Robustness of the Internet. . .

. . . to Deliberate Attack

This block covers

- Modelling the Global Internet Structure
- Network Robustness

Required and Recommended Reading

Required Reading:

- M. E. J. Newman:
  Networks: An Introduction, 2nd ed.
  Oxford University Press, 2018

  Chapter 16 (Percolation and Network Resilience)

Recommended Reading:

- R. Albert, H. Jeong, A.-L. Barabási:
  Error and Attack Tolerance of Complex Networks
  Nature **406**:378–382 (Jul. 2000)
- W. Deng, M. Karaliopoulos, W. Mühlbauer, P. Zhu, X. Lu, B. Plattner:
  k-Fault tolerance of the Internet AS graph
  Computer Networks **55**(10):2492-2503 (Jul. 2011)

Modelling the Internet Structure (1/2)

Global Structural Properties

One of the primary motivations for the development of the Barabási-Albert discussed previously and subsequent models was to obtain more accurate structural network models

- Many different networks from biological to social systems also exhibit scale-free structures, here we are mainly interested in the global Internet structure at (very) large scales
- At different scales, the model is likely not to be accurate: Structured networks will e.g. use deliberate redundancy, avoiding typical hub vertices
- Distributions of nodes may not be *smooth* either

Even where models are fitted to actual data, not all data is actually **visible** — this may lead to inaccurate conclusions and must be considered carefully

Modelling the Internet Structure (2/2)

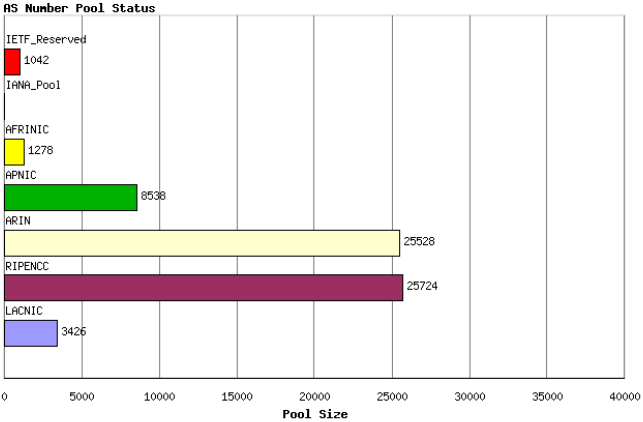
Global Structural Properties

Interconnecting networks at the global level is the role of exterior gateway protocols; at present only one such protocol (the **Border Gateway** (BGP) protocol) is in use in the Internet

- For this, each network requires an unique Autonomous System (AS) number, ultimately assigned by the IANA
  - Originally a 16-bit value, support for 32-bit AS numbers has been in place for several years (RFC 4893, updated by RFC 6793)
- As of 12/2025 there are approx. 78000 advertised active AS (as opposed to allocated) – 1.05m active FIB, 3.11m RIB entries
  - For years vendors marketed core routers as “certified for 1m routes” – this was hit in 2025
  - RIB size is usually not a problem and routers can handle a lot more, FIB prefix size may result in inefficiencies as core routers use TCAM (ternary content-addressable memory) for it
- Each AS must be connected to at least two other AS; apart from the networks in its own organisational unit, it must advertise accessible networks to other ASes

Autonomous System Allocation

16 and 32 Bit AS Statistics

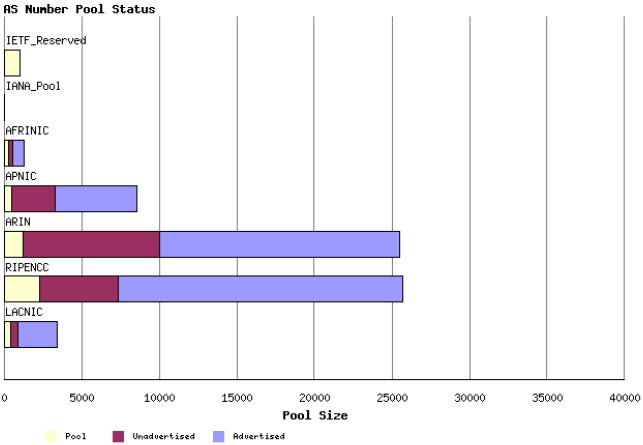


Breakdown of 16-bit AS by regional registries (RIRs) Q2/2020

Source: Geoff Houston

Autonomous System Allocation

16 and 32 Bit AS Statistics

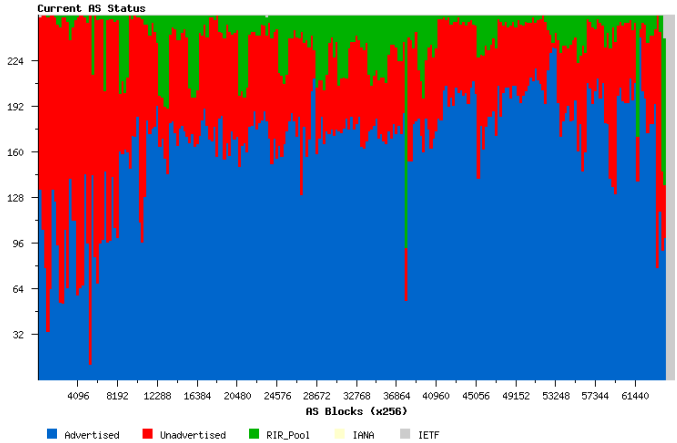


16-bit AS allocation vs advertisement Q2/2020 – now closed

Source: Geoff Houston

Autonomous System Allocation

16 and 32 Bit AS Statistics

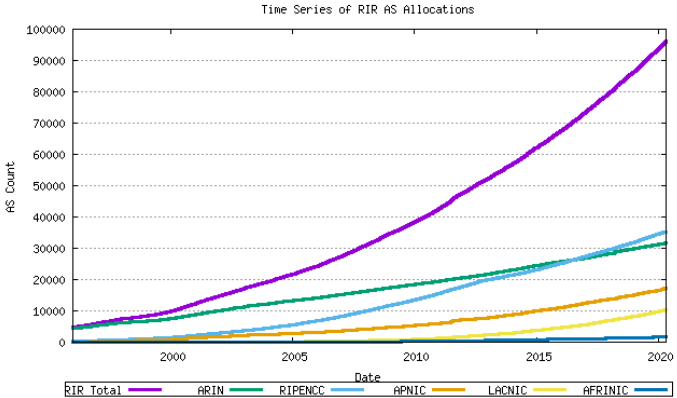


16-bit AS allocation by RIR Q2/2020

Source: Geoff Houston

Autonomous System Allocation

16 and 32 Bit AS Statistics

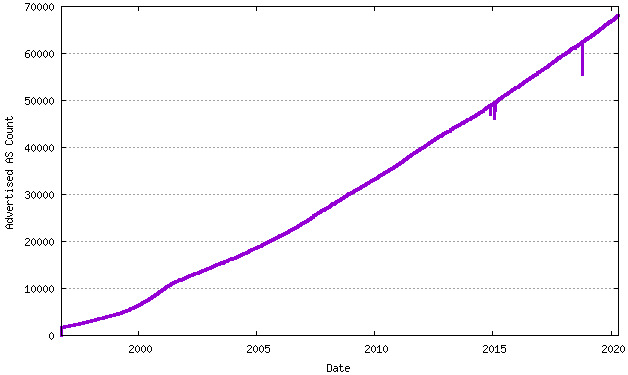


Cumulative RIR AS (32 bit) assignments per RIR 1/1996 to 1/2022

Source: Geoff Houston

Autonomous System Allocation

16 and 32 Bit AS Statistics

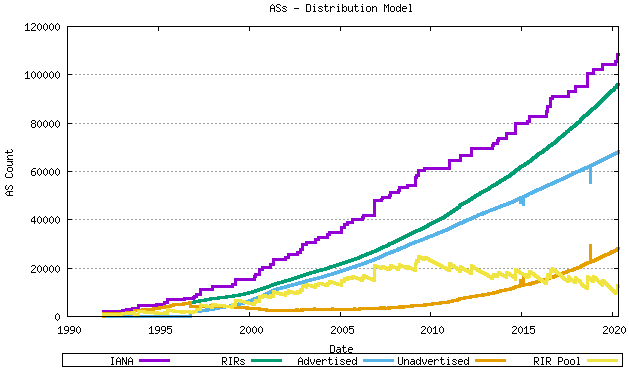


Unique ASN in BGP table 1997–2022

Source: Geoff Houston

Autonomous System Allocation

16 and 32 Bit AS Statistics

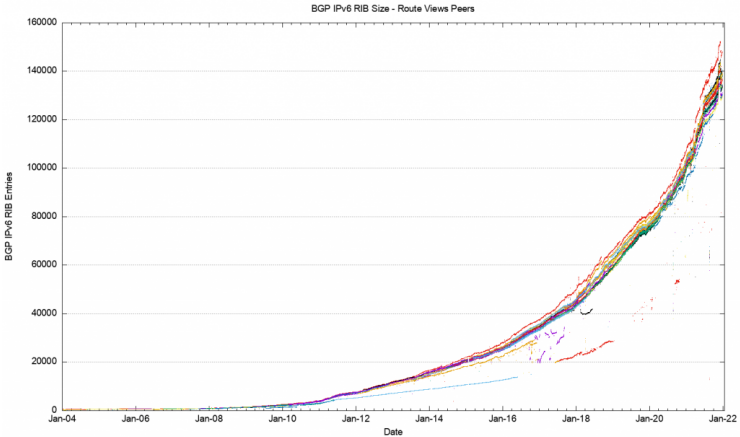


32-bit (cumulative) count 1/1990 to 1/2022

Source: Geoff Houston

Autonomous System Allocation

16 and 32 Bit AS Statistics

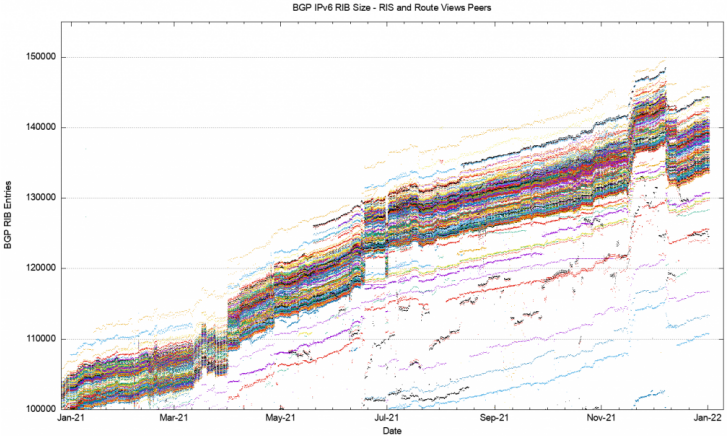


IPv6 RIB size 2004 – 2022

Source: APNIC

Autonomous System Allocation

16 and 32 Bit AS Statistics



IPv6 RIB size 1/2021 – 1/2022: Not Every Router Sees The Same!

Source: APNIC

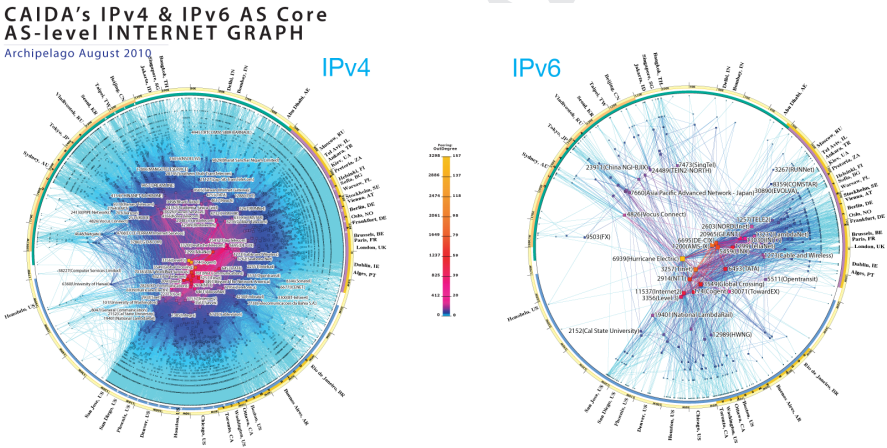
AS Topology

The AS are interconnected by a basic structure:

- Each AS i has a number of border gateways: A router that can transfer messages between AS i and an AS to which that router is *linked*
- If BG_1^i of AS i is linked to BG_1^j of AS $j \Rightarrow$ there exists a direct connection between the two routers
- Two gateways BG_1^i and BG_2^i of the same AS i are always *internally linked*
- A border gateway BG_1^i of AS i, attached to network n_i, announces $\langle i, n_i \rangle$ to its neighbouring gateways
- Assume BG_1^j of AS j is linked to BG_1^i: BG_1^j can then announce that it knows a path to n_i: $\langle j, i, n_i \rangle$
- Each other gateway BG_2^j of AS j can announce $\langle j, i, n_i \rangle$ to *its* linked neighbours

Observed AS-Level Topology

The following are snapshots from August 2010 and January 2014 for both IPv4 and IPv6 BGP topology (cf. 2017 graph shown earlier):



Source: Cooperative Association for Internet Data Analysis (CAIDA, US)

Observed AS-Level Topology

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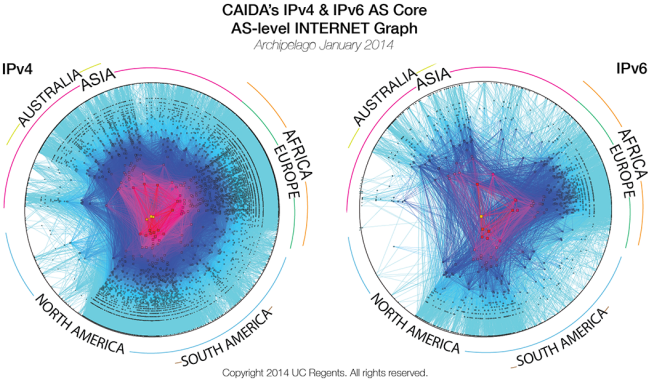
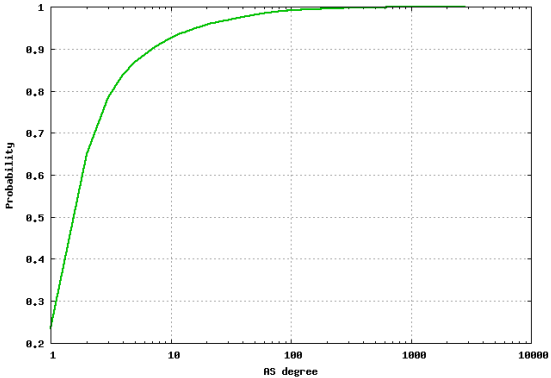


Source: Cooperative Association for Internet Data Analysis (CAIDA, US)

Idealised AS-Level Topology (1)

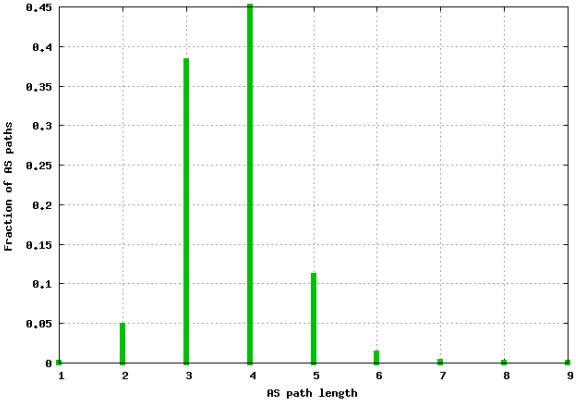
In 2009 (with approximately 30000 AS), the following cumulative degree distribution could be observed for the Internet ignoring policy-based routing decisions:



Source: Internet Research Lab (UCLA) and NEC Research Laboratories (Germany)

Idealised AS-Level Topology (2)

The same data set (with fitting applied) gives the following histogram of path lengths:



Source: Internet Research Lab (UCLA) and NEC Research Laboratories (Germany)

Real AS-Level Topology (1/3)

In reality, commercial and practical considerations cause deviations from the idealised AS adjacency model shown in the previous charts

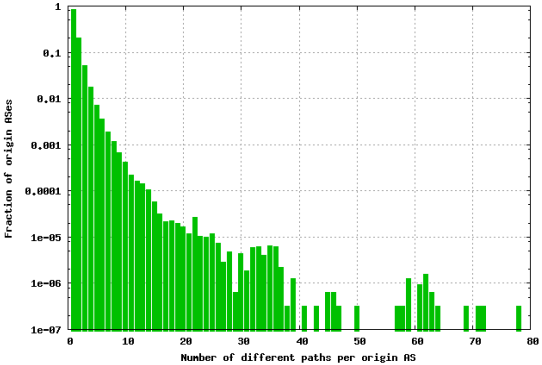
- Decisions at the policy level may also dynamically change this and is not captured in a static model

A popular myth about the Internet is that it was designed to withstand nuclear attacks and the wide-spread devastation caused by these

- Intuitively, one would be able to “route around” any failed component of the network to keep the overall network connected
- In this case one would expect path diversity:
  Paths should be **vertex-disjoint**

Real AS-Level Topology (2/3)

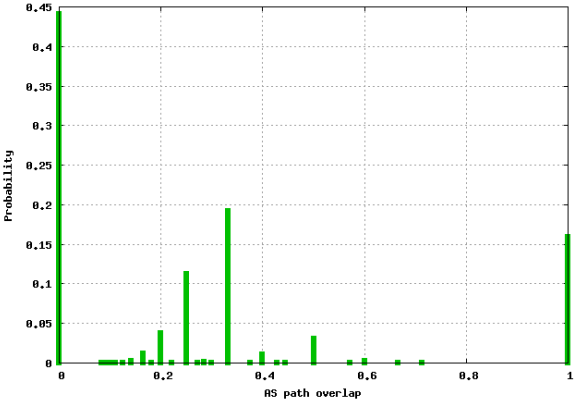
This is clearly not the case as can be seen from this graph showing the fraction of origin ASes with different numbers of AS paths that lead to them (as summed up over the routing tables used):



Source: Internet Research Lab (UCLA) and NEC Research Laboratories (Germany)

Real AS-Level Topology (3/3)

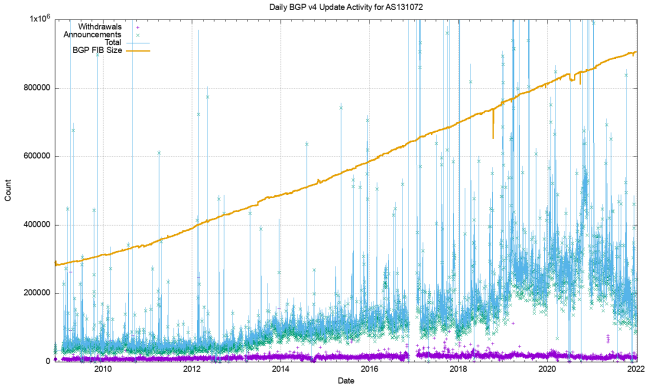
Discrepancy between real and ideal routes (*link overlap* expressed as the fraction of AS-level links that are common to both paths):



Source: Internet Research Lab (UCLA) and NEC Research Laboratories (Germany)

BGP Stability

Prefixes will need to be advertised and withdrawn (legitimately and erroneously):

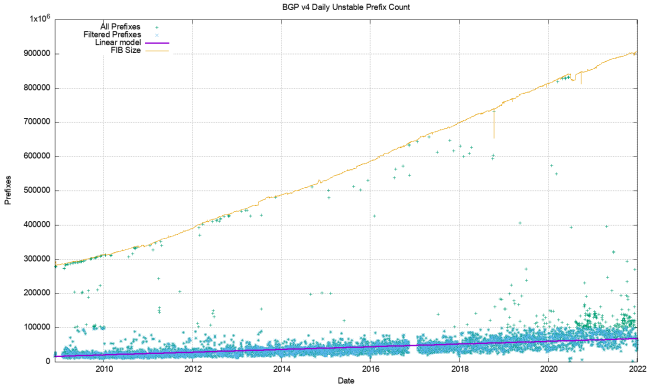


Daily IPv4 BGP Updates 2009–2022

Source: APNIC

BGP Squeaky Wheels

A relatively small number of prefixes is responsible for a large number of updates:

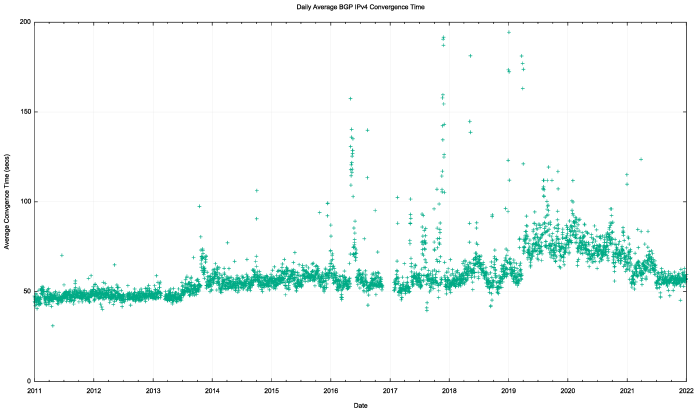


Unstable IPv4 BGP Updates 2009–2022

Source: APNIC

BGP Convergence Time

Convergence times will vary for each BGP speaker, so the following is to be seen only as an indicative trend:



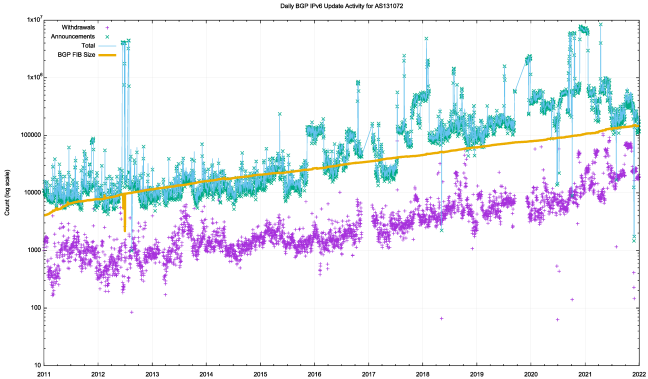
Local BGP Convergence 2009–2022

Source: APNIC

IPv6 BGP Stability

Daily IPv6 Updates

Although one might expect that IPv6 BGP behaviour should be similar to that observed for IPv4, it is rather more noisy:



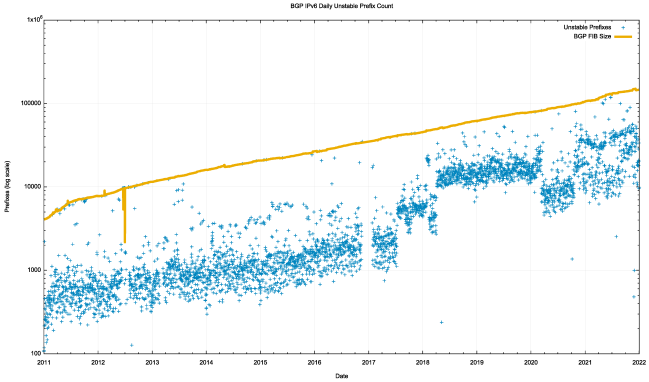
Daily IPv6 BGP Updates 2011–2022

Source: APNIC

IPv6 BGP Stability

Daily Unstable Prefixes in IPv6

Unlike in IPv4, the culprits seem to be more equally distributed:

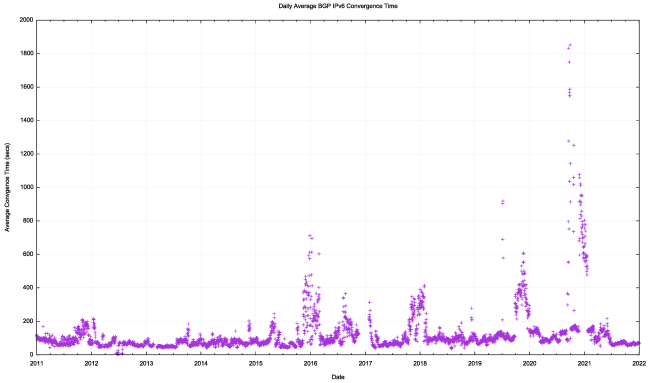


Daily Unstable IPv6 BGP Prefixes 2011–2022

Source: APNIC

IPv6 BGP Stability

IPv6 Convergence Times



Local IPv6 BGP Convergence Times 2011–2022

Source: APNIC

Recurring to Network Models

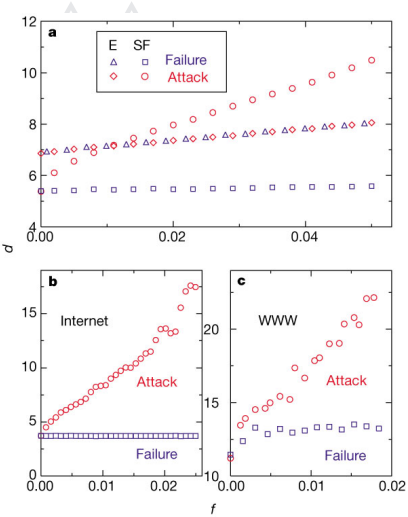
Recall Menger's theorem (from Unit 3) on the number of node-independent paths showing that the number of node-independent paths between two vertices is always equal to the minimum number of vertices in the network that must "fail" for those two vertices to become disconnected from one another

More interesting for us is the global behaviour, i.e. what the effect is if a certain fraction or subset of vertices (or edges) is removed?

- This has been studied intensively, and using a number of different approaches — we can only give examples here
- A key question is the notion of resilience to deliberate attack

Resilience to Attacks
Network Diameter Change under Attack

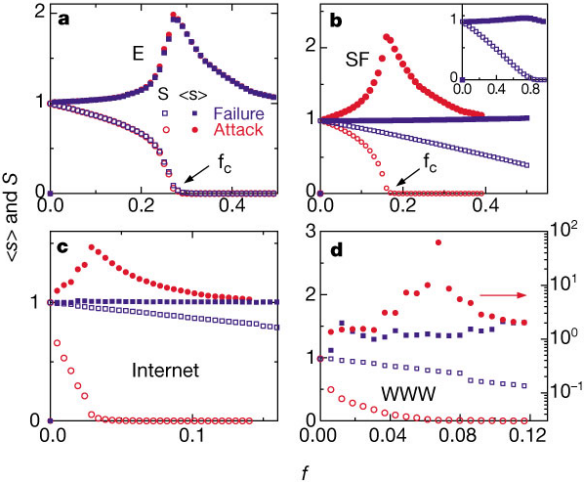
Changes in graph diameter *d* as a function of removed vertex fraction for exponential (E) and scale-free (SF) graphs
Figures (b) and (c) show the effect of random vs. targeted removal of vertex fractions for the AS and WWW connection graphs



R. Albert, H. Jeong, A.-L. Barabási:
Error and Attack Tolerance of Complex Networks Nature **406**:378–382 (Jul. 2000)

Network Fragmentation under Attack

Relative size of largest clusters *S* (open circles) and average size of isolated clusters as a function of the fraction of removed nodes (symbols as on previous slide)



R. Albert, H. Jeong, A.-L. Barabási:
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Robustness to Targeted Attacks

Scale-free graphs are clearly much more resilient to random vertex removal

- This confirms our daily observation that e.g. the Internet continues to function smoothly despite faults, misconfigurations, and arbitrary attacks

Scale-free graphs are **much more susceptible** to targeted attacks

However, recall that the Internet in particular is not strictly a (scale-free) random graph:

- Deliberate structures form part of it, and policy-based routing influences connectivity as shown before

Robustness – Some Remarks

While the results on vulnerability are important, parameters and underlying assumptions must be clearly understood:

- Most models assume a single event affecting robustness, so any repair or re-routing is not necessarily considered
- Where this is the case, or e.g. where flows rather than connectivity are concerned, results can differ significantly — more on this later
- Recent results on temporal robustness for dynamic networks where edges vary indicate that intelligent attacks are more effective than base results for scale-free networks would suggest
- The choice of network parameters can alter results significantly, and these are necessarily relatively crude and uniform as simulations cannot cope with very large networks
- Quite a few updates in BGP seem not driven by reachability but by **traffic engineering**

A Closer Look at Internet Robustness

While the global structure of the Internet (certainly at the AS level) seems to match well with a scale-free model, this does not fully reflect **traffic flows**

- As noted, routing policies must be taken into account in addition to connectivity
- A key property is that unlike the basic BGP model, not all AS are peers: *There is a hierarchical relation among many AS*, with some AS acting as providers
- Non-provider AS do not normally route traffic among peers or even providers and will therefore **not** advertise such edges (routes)
- It is therefore desirable to model relations more fully, by using labels on edges — one possible model is a Type of Relationship (ToR) graph

AS Modelling using ToR Graphs (1/5)

By defining a ToR (directed) graph $G = (V, E, R)$ where $R = \{p2c, c2p, p2p, s2s\}$ identifying provider-customer, customer-provider, peer-to-peer, and sibling-to-sibling relations, a graph modelling relations among AS can be constructed

- In this graph, so-called **Tier 1** AS would need to form a full mesh: This is evident from observations
- Below this tier, multiple *hierarchical* levels of AS can be found, where cross-connections are relatively rare
- Clearly, to obtain resilience to faults, some redundancy will be needed, but the number of edges should be minimised
- Unfortunately in general ToR graphs it is $\mathcal{NP}$-hard to find the maximum number of vertex-disjoint paths (or whether there are k vertex-disjoint valid paths between given nodes) for constant k

AS Modelling using ToR Graphs (2/5)

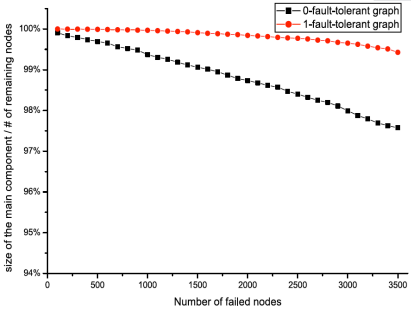
Deng et al. (2011) investigated graph augmentation, but specifically the fault-tolerance properties of basic and augmented AS graphs

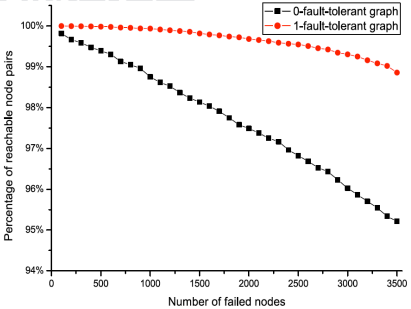
- k fault tolerance informally results in any pair of nodes being able to reach one another with less than k faults
- Apart from computational complexity, k-tolerance for $k \geq 2$ is likely to be considered too expensive
- It is important to note that the resulting augmented graph is still (broadly) a scale-free graph, but is of course no longer *random*

AS Modelling using ToR Graphs (3/5)

Consider n nodes failing ($100 \leq n \leq 3500$), corresponding to approximately 10% of AS nodes

For random failures, the largest connected component (or number of remaining nodes) is shown on the left for both 0- and 1−tolerant ToR graphs on the left, the fraction of reachable node pairs on the right-hand chart:





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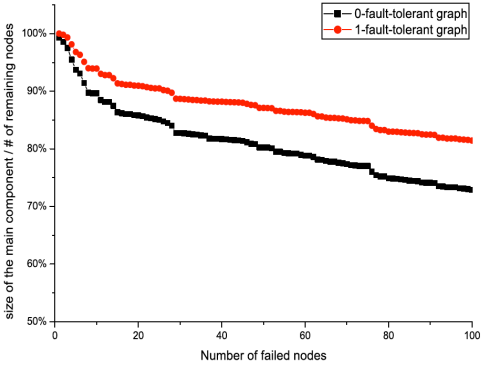
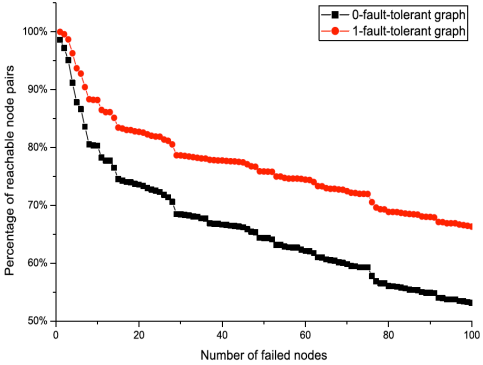
k-Fault tolerance of the Internet AS graph. Computer Networks **55**(10):2492-2503 (July 2011)

AS Modelling using ToR Graphs (4/5)

For deliberate failures (removing vertices with the highest degree first, for $0 \leq n \leq 100$ only), the 1-tolerant graph shows a significant improvement over the familiar effects of deliberate attacks on scale-free graphs:





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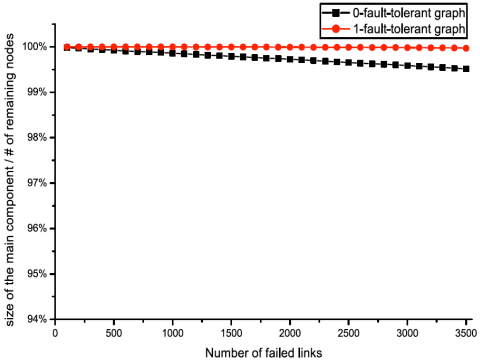
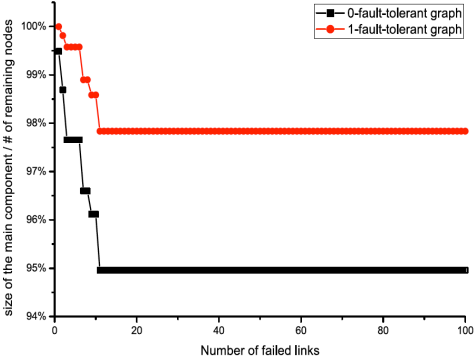
AS Modelling using ToR Graphs (5/5)

Finally, when studying link (edge) removal, we can again see that random link failure has very little impact, while targeted removal of edges with high vertex degree immediately leads to degradation, even in case of the 1-tolerant graph:



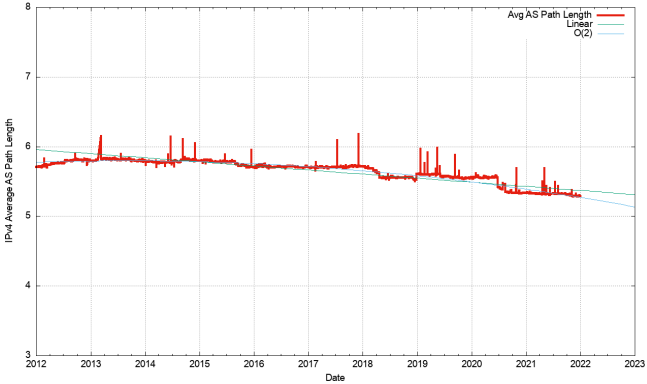


W. Deng, M. Karaliopoulos, W. Mühlbauer, P. Zhu, X. Lu, B. Plattner:
k-Fault tolerance of the Internet AS graph. Computer Networks **55**(10):2492-2503 (July 2011)

IP Convergence Behaviour

IPv4 Average Path Length

Despite explosive growth in the number of nodes, the (here: IPv4) average path length expressed in AS has actually been **shrinking**:



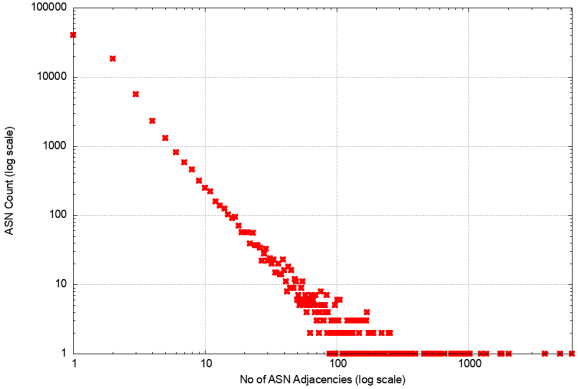
IPv4 AS Average Path Length 2012–2022

Source: APNIC

IPv4 Average Path Length

AS Adjacency

The explanation can be seen in the distribution of AS adjacency in the IPv4 network (again, IPv6 would be expected to be similar, but is actually less stable):



IPv4 AS Adjacency 2012–2022

Source: APNIC

Beyond the Internet Graph

The preceding analysis has only given a limited glimpse into the study of a relatively well-understood network

Network robustness has been investigated under a number of different assumptions for many other types of networks, particularly from critical infrastructures such as the electric power grid

- Questions in these case may extend beyond connectivity, and one may need to study different types of networks
- Consider the example of oil and gas pipelines: Explicit edge and flow models may be required in this case

Summary

This section provided material on

- Modelling the Global Internet Structure
- Network Robustness